LECTURE 18 (153-248)

AR(p):

$$y_t = \phi_0 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

ε_t iid $N(0, \sigma^2)$

parameters: $\phi_0 \ldots \phi_p, \sigma$

① Create $y = \begin{pmatrix} y_{p+1} \\ \vdots \\ y_n \end{pmatrix}$, $X = \begin{bmatrix} 1 & y_p & \cdots & y_1 \\ \vdots & \vdots & & \vdots \\ 1 & y_{n-1} & & y_{n-p} \end{bmatrix}$

$(n-p) \times (p+1)$

→ sm.OLS(y, X)

$\hat{\phi} = (X^T X)^{-1} X^T y$

$\hat{\sigma} = \sqrt{\frac{RSS}{(n-p)-(p+1)}}$

$\hat{\sigma}^2 (X^T X)^{-1}$ → diagonals & take square roots to get standard errors of coefficients

② AutoReg in statsmodels

→ AutoReg(y, lags = p)

$\hat{\phi} = (X^T X)^{-1} X^T y$

$\hat{\sigma}_{ML} = \sqrt{\frac{RSS}{n-p}}$

$\hat{\sigma}^2_{ML} (X^T X)^{-1}$ → square roots of diagonals gives standard errors

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Predictions is a big application of AR models (& related models ARIMA)

$$y_{n+i}(\theta) = \mathbb{E}(y_{n+i} \mid y_1 \ldots y_n)$$

θ

θ : all parameters $(\phi_0 \cdots \phi_p, \sigma^2)$

$$y_{n+1}(\theta) = \mathbb{E}\left[y_{n+1} \mid y_1 \cdots y_n, \theta \right]$$

$$= \mathbb{E}\left[\phi_0 + \phi_1 y_n + \phi_2 y_{n-1} + \cdots + \phi_p y_{n+1-p} + \varepsilon_{n+1} \mid y_1 \cdots y_n, \theta \right]$$

$$= \phi_0 + \phi_1 y_n + \cdots + \phi_p y_{n+1-p} + \underbrace{\mathbb{E}\left(\varepsilon_{n+1} \mid y_1 \cdots y_n, \theta\right)}_{0}$$

$$= \phi_0 + \phi_1 y_n + \cdots + \phi_p y_{n+1-p}$$

$\theta \to \hat{\theta}$ predict y_{n+1} by $y_{n+1}(\hat{\theta})$

$$y_{n+2}(\theta) = \mathbb{E}\left[\phi_0 + \phi_1 y_{n+1} + \phi_2 y_n + \cdots + \phi_p y_{n+2-p} + \varepsilon_{n+2} \mid y_1 \cdots y_n, \theta \right]$$

$$= \phi_0 + \phi_1 y_{n+1}(\theta) + \phi_2 y_n + \cdots + \phi_p y_{n+2-p}$$

$$y_{n+i}(\theta) = \phi_0 + \phi_1 y_{n+i-1}(\theta) + \cdots + \phi_p y_{n+i-p}(\theta)$$

$i = 1, 2, \cdots$

$y_j(\theta) = y_j$ if $1 \le j \le n$

→ Recursion

Replace θ by $\hat{\theta}$.

Autocorrelation

Data: $y_1, y_2, \cdots, y_n$

(y_1, y_2)

(y_2, y_3)

$r :=$ correlation between y_t & y_{t+1}

$$(y_{n-1}, y_n) = \frac{\sum_{t=1}^{n-1}\left(y_t - \bar{y}^{(1)}\right)\left(y_{t+1} - \bar{y}^{(2)}\right)}{\sqrt{\sum_{t=1}^{n-1}\left(y_t - \bar{y}^{(1)}\right)^2}\sqrt{\sum_{t=2}^{n}\left(y_t - \bar{y}^{(2)}\right)^2}}$$

$$\bar{y}^{(1)} = \frac{y_1 + \cdots + y_{n-1}}{n-1}$$

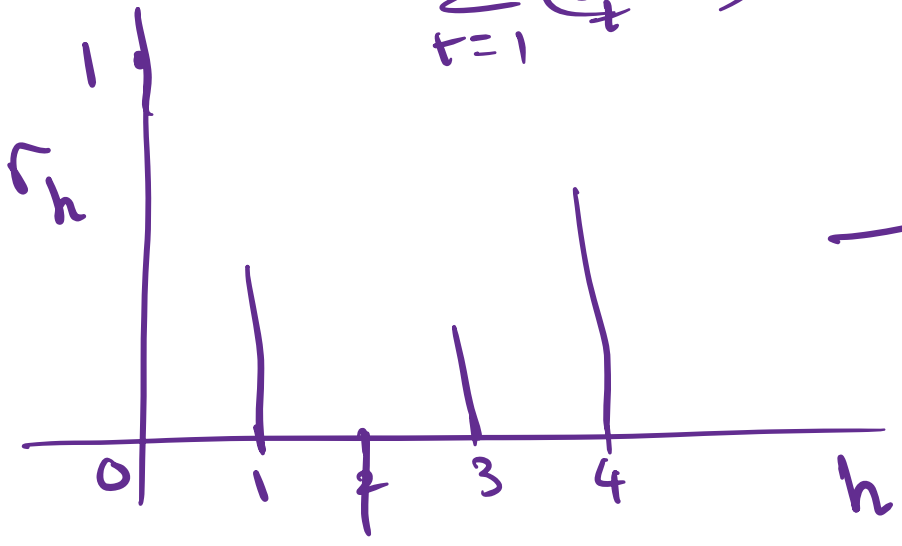
$$\bar{y}^{(2)} = \frac{y_2 + \cdots + y_n}{n-1}$$

both will be similar to $\bar{y}$

$$\approx \frac{\sum_{t=1}^{n-1}(y_t - \bar{y})(y_{t+1} - \bar{y})}{\sum_{t=1}^{n}(y_t - \bar{y})^2}$$

$$r_h = \frac{\sum_{t=1}^{n-h}(y_t - \bar{y})(y_{t+h} - \bar{y})}{\sum_{t=1}^{n}(y_t - \bar{y})^2}$$

} SAMPLE AUTO CORRELATION AT LAG h.



→ Sample ACF plot (AUTO CORRELATION FUNCTION)

plot_acf

MA models:

$\varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$

→ $\varepsilon_t = z_t + \theta z_{t-1}$ where $z_t \overset{iid}{\sim} N(0, \sigma^2)$

Moving Average order 1

$\varepsilon_5 = z_5 + \theta z_4$ & $\varepsilon_6 = z_6 + \theta z_5$

of Orde

$$\varepsilon_t = z_t + \theta_1 z_{t-1} + \theta_2 z_{t-2}, \quad z_t \overset{iid}{\sim} N(0,\sigma^2)$$

MA(2)

$$y_t = \phi_0 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

ε_t: MA(q)

ARMA(p, q)

p=0

$$y_t = \phi_0 + z_t + \theta z_{t-1}$$

$z_t \overset{iid}{\sim} N(0,\sigma^2)$

AR(p) Prediction Uncertainties.

$\theta = (\phi_0, \ldots, \phi_p, \sigma)$

$y_{n+i}(\theta)$

$$V_i(\theta) = \operatorname{var}\left(y_{n+i}(\theta) \,\middle|\, y_1 \ldots y_n, \theta \right)$$

So $\sqrt{V_i(\hat{\theta})}$: prediction standard error for y_{n+i}

$$V_1(\theta) = \operatorname{var}\left(y_{n+1}(\theta) \,\middle|\, y_1 \ldots y_n, \theta \right)$$

$$= \operatorname{var}\left(\phi_0 + \phi_1 y_n + \phi_2 y_{n-1} + \cdots + \phi_p y_{n+1-p} + \varepsilon_{n+1} \,\middle|\, y_1 \ldots y_n, \theta \right)$$

$= \text{var}(\varepsilon_{n+1} \mid \theta) = \sigma^2$

$V_2(\theta) = \text{var}\left(y_{n+2}(\theta) \mid \underset{\theta}{y_1 \cdots y_n} \right)$

$= \text{var}\left(\phi_0 + \phi_1 y_{n+1}(\theta) + \phi_2 y_n + \cdots + \phi_p y_{n+2-p} + \varepsilon_{n+2} \mid \underset{\theta}{y_1 \cdots y_n} \right)$

$= \text{var}\left(\phi_1 y_{n+1} + \varepsilon_{n+2} \mid \underset{\theta}{y_1 \cdots y_n} \right)$

$= \phi_1^2 \underbrace{V_1(\theta)} + \sigma^2 = \sigma^2(1 + \phi_1^2)$

$V_3(\theta) = \text{var}\left(y_{n+3} \mid \underset{\theta}{y_1 \cdots y_n} \right)$

$= \text{var}\left(\underbrace{\phi_1 y_{n+2}} + \phi_2 \underbrace{y_{n+1}} + \underbrace{\varepsilon_{n+3}} \mid \underset{\theta}{y_1 \cdots y_n} \right)$

To compute this, we need to compute $\text{cov}\left(y_{n+2}, y_{n+1} \mid \underset{\theta}{y_1 \cdots y_n}\right)$ also in addition to variances.

Covariance Recursion

$\Gamma_K(\theta) =$

Covariance Matrices

$y_1 \cdots y_n$: n random variables.

$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$: random vector

$$\mathbb{E}Y = \begin{pmatrix} \mathbb{E}Y_1 \\ \vdots \\ \mathbb{E}Y_n \end{pmatrix}$$

$\underbrace{\mathrm{Cov}(Y)}_{n \times n}$ = $n \times n$ matrix with (i,j)-th entry $\mathrm{Cov}(Y_i, Y_j)$

$$\begin{bmatrix} \mathrm{Var}(Y_1) & \mathrm{Cov}(Y_1,Y_2) & \mathrm{Cov}(Y_1,Y_3) & \cdots \\ & \ddots & & \end{bmatrix}$$

① $\mathbb{E}(AY + c)$
$= A(\mathbb{E}Y) + c$

② $\mathrm{Cov}(AY + c)$
$= A\,\mathrm{Cov}(Y)\,A^T$

Y & W are two random vectors

Y: $n \times 1$, W: $m \times 1$

$\mathrm{Cov}(Y, W)$: $n \times m$ matrix whose (i,j)-th entry is $\mathrm{Cov}(Y_i, W_j)$

$$\mathrm{Cov}(AY + b, CW + d) = A\,\mathrm{Cov}(Y, W)\,C^T$$

$$\mathrm{Cov}\left(\begin{pmatrix} y_{n+1} \\ y_{n+2} \\ \vdots \\ y_{n+k} \end{pmatrix} \middle| \; y_1, \ldots, y_n, \theta \right) = \Gamma_k(\theta)$$

$$\Gamma_1(\theta) = \mathrm{var}(y_{n+1} \mid y_1, \ldots, y_n, \theta) = \sigma^2$$

$\Gamma_1(\theta), \ldots, \Gamma_{k-1}(\theta)$

$$\Gamma_k(\theta) = \begin{bmatrix} \Gamma_{k-1}(\theta) & \\ & \end{bmatrix}$$

$$= \begin{pmatrix} \Gamma_{k-1}(\theta) & \begin{matrix} \mathrm{Cov}(y_{n+1}, y_{n+k} \mid y_1 \ldots y_n) \\ \mathrm{Cov}(y_{n+2}, y_{n+k} \mid y_1 \ldots y_n) \\ \vdots \\ \mathrm{Cov}(y_{n+k-1}, y_{n+k} \mid y_1 \ldots y_n) \\ \mathrm{Var}(y_{n+k} \mid y_1 \ldots y_n) \end{matrix} \end{pmatrix}$$